
Corrigé — Exercice 10

1) $u_1 = \frac{1}{3} \cdot 2 + 2 = \frac{8}{3}, \quad u_2 = \frac{1}{3} \cdot \frac{8}{3} + 2 = \frac{26}{9}.$

2a) $v_{n+1} = u_{n+1} - 3 = \frac{1}{3}u_n + 2 - 3 = \frac{1}{3}(u_n - 3) = \frac{1}{3}v_n$: géométrique de raison $\frac{1}{3}$.

2b) $v_0 = u_0 - 3 = -1$, donc $v_n = -\left(\frac{1}{3}\right)^n$ et $u_n = v_n + 3 = 3 - \left(\frac{1}{3}\right)^n$.

3) $\lim_{n \rightarrow +\infty} u_n = 3.$

4) $S_n = \sum_{k=0}^n \left(3 - \left(\frac{1}{3}\right)^k\right) = 3(n+1) - \frac{3}{2}\left(1 - \left(\frac{1}{3}\right)^{n+1}\right),$ et $\frac{S_n}{n} \rightarrow 3.$